Adaptation of Large Vision-Language Models for non-English Languages

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Abstract

The growing popularity of Large Vision-Language Models highlights a key issue with Large Language Models: their training is primarily based on extensive English text corpora. Consequently, the trained models are more prone to error in non-English language generation tasks, even more so in a multimodal setting where task-specific datasets are used. For this reason, Large Language Models need to be adapted to non-English languages, and a coherent multimodal extension must be applied, even in the absence of open and curated resources for these languages. In this work, we explore Large Vision-Language Models adaptation to non-English languages, using machine translation to overcome the lack of curated data. We also study how the evaluation of the results is influenced when training a vision-to-text adapter in different languages. Finally, the importance of using open resources is highlighted, and in line with this philosophy, the codebase of the adaptation pipeline, all trained models and datasets are freely available.

1 Introduction

Large Vision-Language Models (LVLMs) are a specialization of Large Language Models (LLMs), which do not only process textual data but also visual signals. An LVLM can process visual signals either as input or output or both. Throughout this work, we focus on LVLMs that process visual signals as input data. The extension of LLMs to the visual domain allows them to better comprehend our world and solve tasks that they would not be able to understand otherwise, e.g., describing the contents of an image (Wang et al., 2024). In the textual domain, LLMs are usually trained on large data corpora written in English, while in the vision-language domain, the training procedure follows a similar rationale, but the textual data is paired with an image or video as input.

Currently, many resources are openly available to train both LLMs and LVLMs, and there are many text-vision datasets for a wide variety of tasks. However, one core issue of the state-of-the-art is that all these text-vision resources are mostly available only in English. Indeed, even if LLMs are often trained on multilingual data, English is still the most represented language. An example of this is **LLAMA 2** where it represents more than 89.70% of the training data (Touvron et al., 2023) and the authors of the model themselves do not recommend its usage outside of the English language[1]. Furthermore, LVLMs are often trained exclusively in the English language, for example **LLAVA** (Liu et al., 2024b). These models are able to provide some output in another language, but they do not have the same remarkable capabilities they demonstrate for English, and some strategy is required to bridge this gap (e.g. pre-training or adaptation). Another example in the text-vision domain is the **CLIP** model (Radford et al., 2021). CLIP was trained over a large collection of image-text pairs (mainly collected by crawling the internet), using contrastive-learning to learn a shared embedding space for visual and textual embeddings. Therefore, given the training procedure and the data collection methodology, the original CLIP model is not intended for use cases outside of the English language[2]. Some works try to overcome this limitation: Carlsson et al. (2022) developed a multilingual version of CLIP, while Bianchi et al. (2023) fine-tuned a CLIP model to achieve better performance than multilingual CLIP in Italian language tasks.

However, there is a significant lack of resources in terms of data and available pre-trained models for non-English languages both in the textual and

[1] https://huggingface.co/meta-llama/Llama-2-13b-hf#intended-use

[2] https://huggingface.co/openai/clip-vit-base-patch32#out-of-scope-use-cases

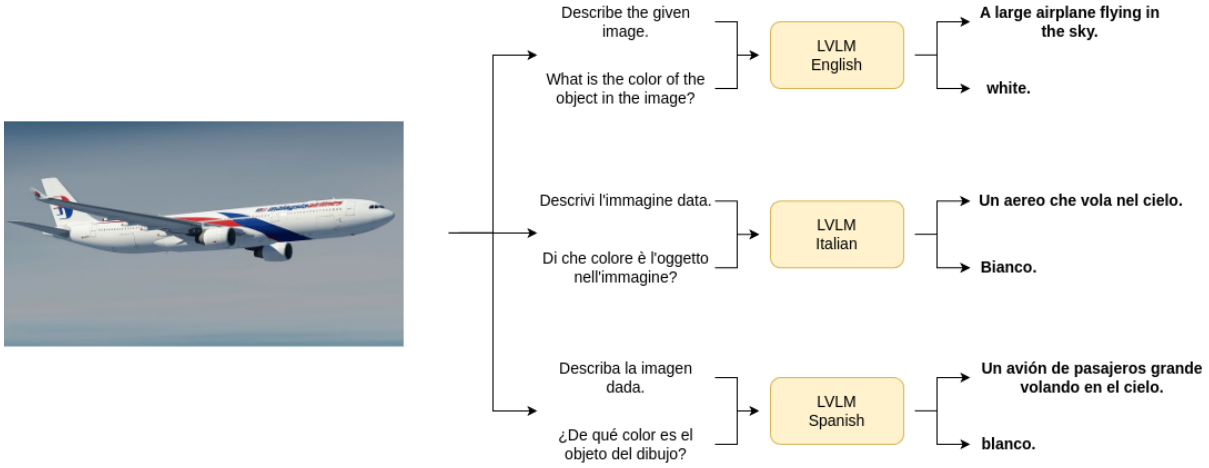


Figure 1: Multi-Task inference example on LVLMs trained on different languages

visual domains. Moreover, the computational cost required to train these models, since they require several GPUs and a long training time to learn effectively, also represents a significant hurdle. Therefore, a procedure that can be easily applied to English becomes difficult to carry out effectively in other languages.

In light of this, we propose a repeatable pipeline for training and evaluating an LVLM in a target underrepresented language. This pipeline encapsulates the whole process of adapting an LVLM to non-English languages, assuming that curated data and powerful models are not available as they are for English. We can summarize our contributions as follows:

- We analyze the effectiveness of training LVLMs for a specific language and compare their performance;
- We investigate the effect of language adaptation on the whole process;
- We release the pipeline to train and evaluate LVLMs for a target language and all the models and datasets obtained from our experiments to further boost research on this topic.

2 Related Works

LVLMs have started seeing widespread usage ever since the release of **GPT-4** (Achiam et al., 2023), the *OpenAI* model which also supported visual input. Following this, many open-source implementations have been developed to extend LLMs to the visual domain in a reproducible and cost-effective way, using a solution that could leverage an already trained LLM. The current methodology mainly involves using an *Adaptation Module* together with the LLM to bridge the modality gap, avoiding a vision-language train from scratch. In the literature, the models that have introduced an adaptation strategy of this type are the following:

- **FLAMINGO** (Alayrac et al., 2022) using a *Perceiver Resampler* which provides a subset of visual tokens obtained from a *Visual Encoder* to *Gated Cross-Attention* layers added to the Language Model;
- **BLIP-2** (Li et al., 2023a) using a *Query Former* module, that is a lightweight Transformer encoder;
- **LLAVA** (Liu et al., 2024b) using a *Projection Matrix* to project visual embeddings into the textual token space.

Some other noteworthy mentions are **INSTRUCTBLIP**, which introduces an *Instruction-aware Query Former* strategy (Dai et al., 2024), and **LLAVA 1.5** (Liu et al., 2024a), which enhances the original by replacing the projection matrix with a *Multi-Layer Perceptron*.

As previously mentioned, a problem with the current state-of-the-art system is the lack of resources for non-English languages. There are some exceptions, typically in the form of multilingual image captioning datasets, but they lack the task diversity that is present in language-vision English datasets. This is fundamental to improve the generalization capabilities of these large models when fine-tuning them using a technique known as *Instruction-Tuning* (Chung et al., 2024). The core

idea is that structuring the input in an instruction-following format helps the models solve unseen tasks better. This methodology is also used for the so-called 'chat' models in which the input format is structured to enhance dialogue capabilities between a user and the model.

As for vision-language datasets that are publicly available and suitable to train these large models, there are **MULTIINSTRUCT** (Xu et al., 2023) and **M3IT** (Li et al., 2023b). Both of them are a collection of publicly available datasets covering a wide range of vision-language tasks. The former consists of 62 tasks (51 used as train split and 11 used as test split) with 5 handwritten instructions for each one, obtained from 21 open-source datasets. The latter consists of 40 tasks with 10 hand-written instructions for each one and also supports multilingual data, providing a machine-translated subset of the original dataset for 80 languages. However, the data quantity is very limited since only 6 tasks are considered for each language, and each task only has around 500 instances.

In the textual domain, adaptation strategies for pre-trained LLMs have been previously developed, Pfeiffer et al. (2020) proposed a framework for cross-lingual transfer called **MAD-X** consisting of three different types of adapters (source language, target language and task) to be applied to pre-trained multilingual models to given languages and tasks; Faisal and Anastasopoulos (2022) extend the previous work by organizing adapters for multiple languages in a hierarchy, to exploit the relationships existing between similar lects.

Finally, regarding the adaptation of vision-language models to non-English languages, some work has been previously done, Shimizu et al. (2018) designed a cross-attention based method between questions and images in the VQA task to map English questions to Japanese ones, exploiting the fact that attention foci of different languages should overlap, Hu and Keller (2023) used a contrastive-loss objective, treating the vision-text alignment as a standalone task. To the best of our knowledge, none of the works in the literature studied vision-language adaptation for LVLMs.

3 Methodology

We propose the following adaptation pipeline for LVLMs:

- **Data Translation**: translation of the data to be used to pre-train or instruction-tune the LVLM;
- **Instruction Formatting**: the translated data is instruction-formatted using pre-defined formats;
- **Language Adaptation**: the pre-trained LLM may be further trained to enhance its knowledge of the target language;
- **Multimodal Adaptation**: an adaptation module, which bridges the gap between the textual and visual modalities, is trained.

For data translation, we distinguish between two different procedures, one done by a human expert and one by a translation model. For the manual translation, one or more human experts, who are proficient in both English and the target language, translate the instruction formats to the target language. This can be done efficiently since the instruction formats are usually static. A machine translation model should be used to translate large-scale data efficiently. This step should also be carefully planned since some tasks may contain textual content that requires an exact match (for example, text extraction tasks from images, which would completely lose their meaning). Therefore, It is important to manually annotate which parts of the data should be subject to automatic translation.

The instruction-formatting step is an automatic procedure that, given the translated data obtained from the previous step, completes the translated instruction formats by adding the machine-translated data to them. Instruction formatting should also add important strings that may be missing in the original instructions (e.g. an `Options` string for close-ended tasks). Finally, additional text may enhance instruction formats to condition model generation (e.g., appending a string which forces the model to answer with a short answer).

As for language adaptation, it is an optional step to boost proficiency of the LLM in the target language. Intuitively, if the model has been pre-trained on a larger quantity of data of a target language, we expect a better performance than a model trained on a collection of predominantly English text. Specifically, LLMs learn by predicting the next token in a sequence given as input. These tokens are obtained by mapping text to tokens learned from a large collection of text by a tokenizer.

Finally, any technique presented in Section 2 could be applied as a multimodal adaptation

methodology. We chose to focus on LLaVA 1.5 (Liu et al., 2024a). Therefore, we will adopt the same multimodal train strategy based on an MLP module. The training procedure consists of two steps: *Alignment* and *Tuning*. In the first step, only the MLP module is trained, while the rest of the architecture weights are kept frozen: the idea is to initialize the MLP module to enhance the next step. In the second step, both the MLP module and the LLM are trained, and only the visual encoder is kept frozen to allow the LLM to learn the new vision-language tasks. Therefore, the traditional LLM auto-regressive objective, that is:

$$P_\theta(x_{i+1}|x_{1:i}) \quad (1)$$

where $x_1, ..., x_i$ are words and θ are the model parameters which encode such probabilities, becomes:

$$P_\theta(x_{m+1}|X, V) \quad (2)$$

where $X = [x_1, ..., x_i]$ and $V = [v_1, ..., v_j]$ such that $m = i + j$. V are visual tokens projected into the textual space by the MLP. With this approach, generation is conditioned on both the textual and visual modalities.

Our intuition is that applying Language Adaptation to the LLM should improve its knowledge of the target language and, therefore, improve its performance after instruction tuning. The idea is that the adaptation step would refine the non-English tokens probabilities of the auto-regressive objective. Optimizing θ for the target language should improve the generation capabilities of the model for that language. Formally:

$$P_{\theta_L}(x_{m+1}|X, V) \quad (3)$$

Where L is the language for which the θ parameters of the model have been optimized. To study whether this assessment is correct, we will be performing experiments with and without language adaptation.

4 Experimental Setup

We apply the proposed pipeline for the Italian and Spanish languages. The model configurations that we study are the following:

- **English**: **ViT CLIP** (Radford et al., 2021) (base, patch size 32) and **LLaMA-2-13B** (Touvron et al., 2023);
- **Italian**: **ViT Italian-CLIP** (Bianchi et al., 2023) (base, patch size 32) and **LLaMAntino-13B** (Basile et al., 2023) (an Italian-adapted LLaMA-2-13B);
- **Spanish**: **ViT Spanish-CLIP**[3] (base, patch size 32) and **LLaMA-2-13B** adapted to the Spanish language.

In our experiments, we tried to use models that were publicly available and for which the language adaptation step had already been done. Since we could not find a language adapted version of LLaMA-2-13B for the Spanish language, we decided to adopt an approach used for causal language modelling also by Conneau and Lample (2019), continuing the pre-training of the model on 1 million articles from a Spanish dump of Wikipedia[4]. We report hyperparameters for the Language Adaptation step in Appendix A. We consider these model architectures for Italian and Spanish and compare them to the English configuration to study the impact of Language Adaptation in the pipeline. Furthermore, the proposed configuration for the English language was never studied in the experiments for LLaVA 1.5. We also experiment with vision encoders extracted from a CLIP adapted model for the target language. Finally, we underline that, in our experiments, we do not aim at creating a language-vision assistant chatbot, differently from LLaVA where the instruction data is enhanced with GPT-4 to have a conversational format. We treat our data as a single-turn conversation where the expected answer is a direct answer that does not require multiple conversation turns. Our aim is to study the reasoning abilities of these models when tuned over a diverse and rich set of tasks.

4.1 Dataset

Following the LLaVA train methodology, we employ two datasets, one for pre-training and one for instruction-tuning.

For pre-training, we use the same dataset as LLaVA 1.5, that is **LCS-558K**[5] , a subset of the **LAION/CC/SBU** dataset paired with BLIP synthetic captions.

[3] https://huggingface.co/flax-community/clip-vit-base-patch32_marian-es

[4] https://huggingface.co/datasets/wikimedia/wikipedia/viewer/20231101.es

[5] https://huggingface.co/datasets/liuhaotian/LLaVA-Pretrain

For Instruction-Tuning, we used MultiInstruct (Xu et al., 2023). We retrieve its train and test splits using the pipeline that is available on GitHub[6]. We also remove the instances with no target text (empty answer). After performing this step, the complete train and test splits consist of $563,863$ (out of 51 train tasks) and $178,617$ (out of 11 test tasks) instances, respectively. For each of these instances, one of the 5 possible instruction formats provided by the dataset for each task is chosen at random.

Note that we do not use the same processing and special tokens used by Xu et al. (2023) in their experiments since they were following the design goals of **OFA** (Wang et al., 2022), which train a model under a unified vocabulary for all modalities.

We normalize the bounding box values to the [0, 1] range and apply the same processing operations that are used for the images in LLaVA 1.5 processing (i.e. padding to keep the aspect ratio, resize, and centre crop) so that the values are consistent with the reference images and independent of their resolution. We append a special `<image>` token randomly either at the beginning or at the end of the instruction (this token is replaced during train and inference by the visual tokens obtained from the MLP module).

Additionally, for tasks that provide options to choose from within the instruction, we format them as an ordered list using either numbers, uppercase or lowercase letters. When doing so, we also replace the target text to predict with the corresponding option identifier. To guide the model's responses, we append special strings before the instruction: *"Answer with the option's number from the given choices directly."* when the options are identified by numbers, *"Answer with the option's letter from the given choices directly."* when the options are identified by letters, *"Answer using a bounding box."* when the target text is a bounding box and, finally, *"Answer the question using a single word or phrase."* for all other cases. We append the corresponding translation to Italian and Spanish, obtained as described in Section 4.2.

Since the data from the MultiInstruct tasks is only in English, we translate it following the methodology described in the next section.

4.2 Translation

For the automatic translation of the data, we use **MADLAD400-3B-MT** (Kudugunta et al., 2024), one of the latest machine-translation models based on a T5 architecture (Raffel et al., 2020). We use the checkpoint available on Huggingface[7].

Additionally, for the Open and Closed VQA tasks, where the answer strictly depends on the context of the question, we concatenate question and answer strings before translating them and we extract the corresponding output for both fields through a regex matching operation.

We also automatically translate the pre-train dataset and the BLIP-generated captions using the same approach.

For the manual translation, we are helped by a human expert proficient in English and the target language (one for Italian and one for Spanish). The experts manually translate the MultiInstruct instruction formats, the pre-train dataset prompts, and the strings to append to the data.

4.3 Training

We use the model configurations presented in Section 4, apply no formatting during pre-training and apply the 'mistral' instruction-format (Jiang et al., 2023) during instruction-tuning. We focus on the 13B version of the model since Liu et al. (2024a) found that scaling up the model was a key aspect of improving performance. For the hyperparameters, we closely follow the ones used by LLaVA 1.5 for both pre-training and instruction-tuning, which we also report in Appendix A.

Listing 1: Mistral Single-Round Instruction template

```
<s>[INST] Instruction [/INST] Model answer</s>
```

Training is performed on a cluster where each node is equipped with 4 NVIDIA A100 64 GB GPUs. We used 2 nodes for a total of 8 GPUs. Alignment and tuning for the multimodal adaptation step take $\sim$ 2 hours and $\sim$ 24 hours, respectively.

4.4 Evaluation

We evaluate the models to check the following criteria:

- Are the models capable of appropriately answering the vision-language tasks of their main language?
- Are the models capable of appropriately answering vision-language tasks of a different language?

[6] https://github.com/VT-NLP/MultiInstruct

[7] https://huggingface.co/google/madlad400-3b-mt

Task	Metric	Model				
		EN-LLaMA EN-CLIP EN-DS	EN-LLaMA EN-CLIP IT-DS	IT-LLaMA IT-CLIP IT-DS	EN-LLaMA EN-CLIP ES-DS	ES-LLaMA ES-CLIP ES-DS
Commonsense VQA	ACC	48.40	48.73	48.79	**50.08**	42.54
Grounded VQA	ACC	**53.65**	53.04	52.33	53.38	47.82
Disaster Damage Severity	ACC	**56.02**	40.74	39.08	43.10	47.94
Disaster Type Classification	ACC	**43.84**	39.06	33.35	36.24	35.20
Disaster Informative	ACC	64.61	65.33	65.58	**66.98**	55.87
NLVR	ACC	**54.00**	51.06	49.14	53.66	53.69
Text VQA	ROUGE-L	**16.79**	13.12	12.35	12.92	4.97
Visual Dialog	ROUGE-L	**36.88**	31.02	33.52	31.54	24.41
Visual NLI	ACC	43.47	43.46	43.00	**43.95**	40.44
Visual Spatial Reasoning	ACC	52.71	52.91	53.48	**54.11**	52.14
Visual Text Extraction	ROUGE-L	**29.13**	17.82	14.87	13.61	1.28

(a) Results on the English test set

Task	Metric	Model				
		EN-LLaMA EN-CLIP EN-DS	EN-LLaMA EN-CLIP IT-DS	IT-LLaMA IT-CLIP IT-DS	EN-LLaMA EN-CLIP ES-DS	ES-LLaMA ES-CLIP ES-DS
Commonsense VQA	ACC	45.07	44.82	44.61	**45.82**	38.27
Grounded VQA	ACC	49.59	**52.04**	51.41	50.36	44.57
Disaster Damage Severity	ACC	45.58	39.63	39.62	**46.47**	42.39
Disaster Type Classification	ACC	52.91	52.81	53.02	**55.12**	46.98
Disaster Informative	ACC	64.96	67.37	66.60	**70.34**	55.12
NLVR	ACC	**53.59**	51.77	50.60	52.77	50.86
Text VQA	ROUGE-L	10.36	**13.96**	11.57	10.86	4.53
Visual Dialog	ROUGE-L	22.03	34.09	**35.13**	28.99	26.99
Visual NLI	ACC	41.91	**42.88**	42.67	39.22	36.58
Visual Spatial Reasoning	ACC	51.84	**53.81**	53.56	53.45	51.84
Visual Text Extraction	ROUGE-L	**32.75**	6.34	4.36	6.64	0.07

(b) Results on the Italian test set

Task	Metric	Model				
		EN-LLaMA EN-CLIP EN-DS	EN-LLaMA EN-CLIP IT-DS	IT-LLaMA IT-CLIP IT-DS	EN-LLaMA EN-CLIP ES-DS	ES-LLaMA ES-CLIP ES-DS
Commonsense VQA	ACC	45.42	44.89	44.38	**46.12**	38.70
Grounded VQA	ACC	51.71	51.62	50.80	**53.16**	47.12
Disaster Damage Severity	ACC	50.16	54.06	54.42	**54.55**	51.45
Disaster Type Classification	ACC	48.74	48.33	**50.51**	46.61	48.07
Disaster Informative	ACC	65.39	66.07	66.08	**69.35**	55.55
NLVR	ACC	**53.79**	49.90	51.47	52.63	52.87
Text VQA	ROUGE-L	10.52	**12.26**	10.75	12.10	5.50
Visual Dialog	ROUGE-L	24.43	31.27	31.07	**37.44**	34.25
Visual NLI	ACC	42.98	42.93	42.11	**44.43**	41.11
Visual Spatial Reasoning	ACC	**54.61**	51.81	52.17	54.28	52.10
Visual Text Extraction	ROUGE-L	**31.77**	10.00	5.88	3.23	0.06

(c) Results on the Spanish test set

Table 1: Results obtained for all test sets, the configurations are formalized as follows: <LANG>-LLAMA where <LANG> is the main language of the LLaMA 2 model, <LANG>-CLIP where <LANG> is the main language of the CLIP model from which the vision encoder is extracted, <LANG>-DS where <LANG> is the main language of the train dataset. <LANG> is always represented using ISO 639-1 language codes. For each task, the best result is reported in bold

Model	Test Set Language		
	English	Italian	Spanish
EN-LLaMA + EN-CLIP + EN-DS	**25.11**	3.09	2.68
EN-LLaMA + EN-CLIP + IT-DS	2.53	**18.45**	5.82
IT-LLaMA + IT-CLIP + IT-DS	3.75	**18.29**	5.89
EN-LLaMA + EN-CLIP + ES-DS	0.54	8.04	**21.52**
ES-LLaMA + ES-CLIP + ES-DS	1.48	6.66	**17.13**

Table 2: Results for Open-ended Crossmodal-3600 image captioning using ROUGE-L. For each model, the best result is reported in bold

Model	Test Set Language		
	English	Italian	Spanish
EN-LLaMA + EN-CLIP + EN-DS	66.66	65.93	**67.24**
EN-LLaMA + EN-CLIP + IT-DS	**69.25**	64.00	67.49
IT-LLaMA + IT-CLIP + IT-DS	**69.59**	65.26	68.33
EN-LLaMA + EN-CLIP + ES-DS	**66.43**	64.73	64.90
ES-LLaMA + ES-CLIP + ES-DS	**68.16**	66.10	64.96

Table 3: Results for Close-ended Crossmodal-3600 image-text selection using Accuracy. For each model, the best result is reported in bold

For visual-language tasks, we follow the evaluation methodology used by Xu et al. (2023), studying the performance on 11 tasks that are not in the train set. All tasks have been translated using the methodology described in Section 4.2, except for *Visual Text Extraction*, which is an optical character recognition task (extracting the text that is present in the given image) and is kept in the English language for all test sets (only the instruction language changes depending on the test set). We use **ROUGE-L** (Lin, 2004) and **Accuracy** as metrics for the tasks in the dataset test set, since they are the same metrics used by Xu et al. (2023). In particular, we use Accuracy for tasks which provide options to choose from (testing if the model predicts the correct option) and ROUGE-L for tasks where no options are provided and the model has to generate a somewhat complex string (e.g. extracting the text from an image). For ROUGE-L, we use the implementation provided in the *datasets*[8] library, in particular we report the *F1* scores. The inference procedure is done in a zero-shot setting, meaning that no examples of input-output are provided to the model in the prompt, and greedy decoding is used as the generation strategy. We report the evaluation results for the English, Italian and Spanish test sets in Table 1. According to the results, tasks evaluated using Accuracy show similar performance across the models for each test set. To study whether these performance differences are significant, we perform **McNemar's test** (McNemar, 1947) on the results of tasks evaluated using Accuracy. We treat the classification results between two systems as a 2x2 contingency table where each cell encodes a possible combination of the outcomes between the two systems. Following best practices, we use a chi-squared test with continuity correction when the sum of the cases where the two classifiers predict a different option is more than or equal to 25 and an exact binomial test otherwise. We consider comparing both the test set (using the same model) as well as the model configurations (under the same test set). The results of this process confirm that for most combinations, there is statistical significance (that is, a p-value lower than 0.05) out of the 8 accuracy-evaluated tasks. The only combinations for which there is a low rate of statistical significance are the Italian ones. We will justify these results in Section 4.4.2. Complete results can be found in Appendix A.

However, given that all models perform well across all test sets independently of the language, we argue that Close-ended tasks are not appropriate for evaluation in this scenario. That is because the model only has to generate the option identifier independently from the test set's language. These models should instead be evaluated on Open-ended generation tasks, where they need to generate text in the same language as the test set. The results of the *Visual Dialog* task, for example, hint to this conclusion, since it is an Open-ended generation task (translated to the target language) and the results are different between the models and test sets.

4.4.1 Analysis of Open-ended Evaluation

To further compare the effects of Open-ended evaluation against Close-ended, we setup an experiment with *Crossmodal-3600* (Thapliyal et al., 2022). Differently from the MultiInstruct test set, the dataset is manually translated into several languages (therefore, there are no artefacts generated by the machine translation model). The dataset consists of 3,600 images paired with human-written captions. Since each image can have more than one caption, we always select the first one in the dataset. We setup two experiments, an Open-ended one where the model has to generate a caption (*image captioning* task) and a Close-ended one where the model has to select the correct caption from a list of provided options (*image-text selection* task). For the Close-ended evaluation, we select 3 differ-

[8] https://github.com/huggingface/datasets

Model	Test Set Language		
	English	Italian	Spanish
EN-LLaMA + EN-CLIP + EN-DS	**22.56**	21.26	20.31
EN-LLaMA + EN-CLIP + IT-DS	62.09	**14.85**	100.83
IT-LLaMA + IT-CLIP + IT-DS	62.5	14.92	168.67
EN-LLaMA + EN-CLIP + ES-DS	123.99	156.43	**15.15**
ES-LLaMA + ES-CLIP + ES-DS	1999.88	563.72	18.90

Table 4: Results for Open-ended Crossmodal-3600 image captioning using Perplexity. For each language, the best result is reported in bold

ent random captions from the dataset and combine them with the correct one to obtain the options list. We instruction-format the dataset according to the tasks. Note that both the image captioning and the image-text selection tasks are also present in the train set (using different datasets). Evaluation results are shown in Table 2 and 3 for the image captioning and image-text selection tasks respectively. The models score higher for image captioning when evaluated on test sets in the same language as their train set. Furthermore, we can confirm that the models perform well across all test sets in image-text selection and even have the best results for test sets different from their target language.

However, we noticed that the results for Open-ended evaluation between language-adapted and multimodal adapted-only models are close. Because of this, we also decided to study the impact of language adaptation in the next sub-section.

4.4.2 Analysis of Language Adaptation

Finally, to assess if the step of language adaptation helped the LVLMs during the generation, we evaluate the models on *Crossmodal-3600* image captioning using Perplexity. Perplexity is a metric that allows to understand how certain a model is of the next token to generate, therefore it allows to study the natural language distribution a model has learned. Perplexity is difficult to compare between different LLMs; however, in this scenario, given that the models share the same architecture, adaptation module and tokenizer, it can be safely used as a comparison metric between models evaluated using a test set of a fixed language (since tokenization will always result in the same number of tokens across the models).

We compute perplexity by instruction-formatting the dataset for the image captioning task and then computing the exponential of the cross-entropy loss obtained on the tokens of the model's response only (the image caption). Results are shown in Table 4.

The obtained results match the ones obtained in the Open-ended evaluation done in Section 4.4.1. Furthermore, we can also gain a deeper understanding of the linguistic knowledge of these models. An interesting finding is that the Spanish configuration (ES-LLAMA + ES-CLIP + ES-DS) has almost completely forgotten the English and Italian languages. This is confirmed by the very high perplexity obtained for that configuration on these test sets. Not only that, but it actually has higher perplexity than the English-only configuration. This suggests that knowledge is language-independent, and specializing models for a given language limits their reasoning capabilities. We also highlight that this phenomenon did not happen for the Italian configuration, however the Italian-adapted LLM was trained using LoRA (which has been found to limit the degree of forgetting of LLMs Biderman et al., 2024), while our Spanish-adapted LLM has been trained using a full-parameter paradigm.

5 Conclusions and Future Works

In this work, we designed and tested an adaptation pipeline for LVLMs in order to create a vision-language LLM-based model for a target non-English language. The pipeline consists of four separate steps: Data Translation, Instruction Formatting, Language Adaptation and Multimodal Adaptation. We present several experiments on the English, Italian and Spanish languages where this pipeline is applied. We also study the effect of language adaptation, using state-of-the-art datasets and models for all steps. We use the MultiInstruct dataset for training, which consists of 51 rich and diverse train tasks, making our models some of the most diverse in the current literature. We evaluate the 11 test tasks of the MultiInstruct dataset and study the evaluation results. We also further studied the role of Open-ended tasks against Close-ended ones and language adaptation's impact on the models. We release our codebase at https://anonymous.4open.science/r/5D7D/.

In future works, we plan to apply this pipeline to newer and more powerful architectures (e.g. LLaMA 3 or Mixtral) and further study best training and evaluation practices for language-adapted LVLMs. In particular, we plan to compare multilingual against unilingual adaptation.

Ethics Statement

The training procedure is performed on the base version of the LLaMA-2 13B models, which did not undergo any safety testing. The datasets used for pre-training and instruction-tuning also did not undergo any such procedure. Therefore, the trained models may produce inaccurate or biased results.

Some of the datasets used for instruction-tuning from MultiInstruct require the acceptance of some legal conditions before downloading. Therefore, as is also done in the MultiInstruct pipeline, we do not re-distribute them and do not provide an automatic download procedure for such datasets.

Human annotators who helped us translating the data participated voluntarily. We explained to them how the data would be used and our interest in this process.

Figure 1 presents an example of inference using an image. The image is in the public domain and obtained from Wikimedia[9].

Limitations

When formatting the dataset for close-ended tasks, we used an identifier for each option (number or letter). We did not use a natural format where the answer would be the text of the option itself.

This work is done using the Italian and Spanish languages, which have a reasonable amount of available resources, making it possible to use machine translation models with ease. However, for even more underrepresented languages, the pipeline that we studied may not be suitable for use, as even machine translation models may not be available due to the scarcity of resources.

Furthermore, the use of a machine translation model could represent a limitation since it might cause artefacts in the translation process. We relied on this approach mainly due to the lack of large-scale human-curated resources for Italian and Spanish.

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A Appendix

We provide additional information regarding our experiments. Table 5 reports hyperparameters used for language adaptation of LLaMA-2, while Table 6 shows the ones used for multimodal adaptation (matching the ones of LLaVA 1.5).

In Table 7, we report the number of times that we found statistical significance between model configurations out of the 8 accuracy-evaluated tasks of the MultiInstruct test set, while in Table 8, we do the same but comparing results obtained on the same model across different test sets.

Table 9 reports pre-train instructions used during the alignment of the MLP for English (originally

Parameter	Language Adaptation
batch size	1024
lr	1e-4
lr schedule	cosine decay
lr warmup ratio	0
weight decay	1e-4
epoch	1
optimizer	AdamW
DeepSpeed stage	3

Table 5: Language Adaptation Hyperparameters

Parameter	Pre-Train	Instruction-Tuning
batch size	256	128
lr	1e-3	2e-5
lr schedule	cosine decay	cosine decay
lr warmup ratio	0.03	0.03
weight decay	0	0
epoch	1	1
optimizer	AdamW	AdamW
DeepSpeed stage	2	3

Table 6: Multimodal Adaptation Hyperparameters

provided by the MultiInstruct authors), Italian and Spanish.

The diagram in Figure 2 summarizes the proposed pipeline. It starts with data collection of the English datasets, specifically the INSTRUCTION FORMATS, the MULTIMODAL TASK CORPUS (e.g. *image captioning*, *vqa*, ...) and the PRE-TRAIN MULTIMODAL CORPUS (to align the MLP of the LLaVA architecture). The data is then translated by either a HUMAN TRANSLATOR, for the instruction formats, or by a MACHINE TRANSLATOR, for the rest. Once the translated data is obtained, the instruction-formatted data is obtained by using a FORMATTER, which applies the TRANSLATED TASK DATA to the proper fields of the TRANSLATED INSTRUCTION FORMATS. After this step, the data processing is completed and it is possible to perform multimodal adaptation. Before this, the LLM can optionally undergo a LANGUAGE ADAPTATION process. The multimodal adaptation process begins with the PRE-TRAIN of the MLP module using the TRANSLATED PRE-TRAIN DATA. Once the PRE-TRAINED MLP is obtained, the final step of INSTRUCTION-TUNING on the TRANSLATED INSTRUCTION FORMATTED TASK DATA is carried out to obtain the final model: the TARGET LANGUAGE LLAVA.

Test Set	Model 1	Model 2			
		EN-LLaMA EN-CLIP EN-DS	EN-LLaMA EN-CLIP IT-DS	IT-LLaMA IT-CLIP IT-DS	EN-LLaMA EN-CLIP ES-DS
English	EN-LLaMA EN-CLIP IT-DS	4	-	-	-
	IT-LLaMA IT-CLIP IT-DS	5	4	-	-
	EN-LLaMA EN-CLIP ES-DS	6	7	7	-
	ES-LLaMA ES-CLIP ES-DS	6	6	7	7
Italian	EN-LLaMA EN-CLIP IT-DS	4	-	-	-
	IT-LLaMA IT-CLIP IT-DS	4	2	-	-
	EN-LLaMA EN-CLIP ES-DS	6	6	6	-
	ES-LLaMA ES-CLIP ES-DS	7	7	6	7
Spanish	EN-LLaMA EN-CLIP IT-DS	5	-	-	-
	IT-LLaMA IT-CLIP IT-DS	6	4	-	-
	EN-LLaMA EN-CLIP ES-DS	6	6	6	-
	ES-LLaMA ES-CLIP ES-DS	7	7	6	7

Table 7: Statistical significance results between different test sets for the same model configuration using McNemar's test.

Model	**Test Set 1**	**Test Set 2**	
		English	Italian
EN-LLaMA EN-CLIP + EN-DS	Italian	5	-
	Spanish	6	5
EN-LLaMA EN-CLIP + IT-DS	Italian	6	-
	Spanish	6	4
IT-LLaMA IT-CLIP + IT-DS	Italian	4	-
	Spanish	6	4
EN-LLaMA EN-CLIP + ES-DS	Italian	6	-
	Spanish	5	5
ES-LLaMA ES-CLIP + ES-DS	Italian	6	-
	Spanish	5	4

Table 8: Statistical significance results between different model configurations for the same test set using McNemar's test.

English	Italian	Spanish
What is this?	Che cos'è questo?	¿Qué es?
Render a clear and concise summary of the photo.	Fornisci un riassunto chiaro e conciso della foto.	Hacer un resumen claro y conciso de la foto.
Present a compact description of the photo's key features.	Presenta una descrizione compatta delle caratteristiche principali della foto.	Presentar una descripción compacta de las principales características de la foto.
Describe the image concisely.	Descrivi l'immagine in modo conciso.	Describir la imagen de forma concisa.
Write a terse but informative summary of the picture.	Scrivi un riassunto sintentico ma informativo dell'immagine.	Escribir un resumen conciso y informativo de la image.
What is in the photo?	Che cosa c'è nella foto?	¿Qué hay en la foto?
Provide a brief description of the given image.	Fornisci una breve descrizione dell'immagine data.	Proporcionar una breve descripción de la imagen dada.
Give a short and clear explanation of the subsequent image.	Dai una spiegazione breve e chiara dell'immagine che segue.	Dar una explicación breve y clara de la imagen siguiente.
Give a brief description of the image.	Dai una breve descrizione dell'immagine.	Describir brevemente la imagen.
Summarize the visual content of the image.	Riassumi il contenuto visivo dell'immagine.	Resumir el contenido visual de la imagen.
Share a concise interpretation of the image provided.	Condividi un'interpretazione concisa dell'immagine fornita.	Ofrecer una interpretación concisa de la imagen proporcionada.

Table 9: Pre-Train instructions in English, Italian and Spanish.

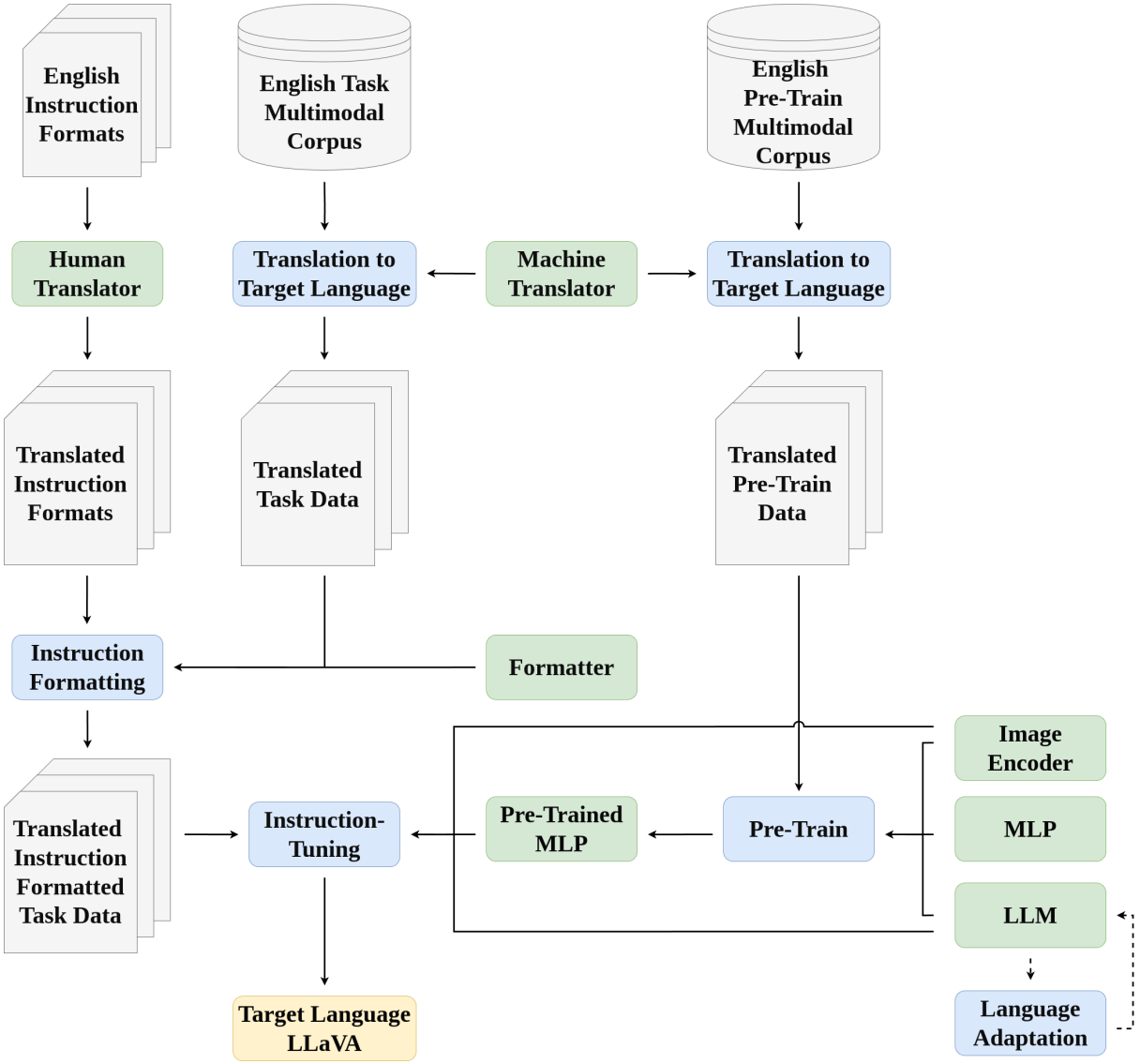


Figure 2: Adaptation pipeline using LLaVA as LVLM.